

Performance Testing

Date	15 July 2026
Team ID	
Project Name	Book A Doctor
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. The sections below document the testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing, Stress Testing
Target Module / API	Appointment Booking API, Login API, Payment Gateway
Test Environment	Cloud-hosted server (AWS EC2, 8 GB RAM, 4 vCPUs)
Test Date	12 July 2026

Step 2: Test Scenarios

S.No	Metric	Target Value	Actual Value	Status	Remarks
S.No	Test Scenario / Description		No. of Virtual Users	Duration (sec)	Expected Outcome

1	100 users booking appointments simultaneously	100	120	System handles requests without timeout
2	500 users logging in at peak hours	500	180	Login API responds within 2 seconds
3	Sudden spike of 300 users making payments	300	90	Payment gateway processes without failure
4	Continuous load of 200 users searching doctors	200	300	Search results delivered under 2 seconds

Step 3: Performance Test Results

1	Response Time (Avg)	< 2 seconds	1.8 seconds	Pass	Within acceptable range
2	Response Time (Max)	< 5 seconds	4.2 seconds	Pass	Slight spike under heavy load
3	Throughput (Req/sec)	200 req/sec	220 req/sec	Pass	Higher than expected
4	Error Rate	< 1%	0.6%	Pass	Minor errors during payment spike
5	CPU Utilization	< 80%	72%	Pass	Stable under load
6	Memory Utilization	< 80%	68%	Pass	No memory leaks observed

Step 4: Observations & Analysis

Key Findings:

The system performs well under both normal and peak load conditions.
 Appointment booking and login APIs are stable and responsive.
 Payment gateway integration handled spikes with minimal errors.

Bottlenecks Identified:

Slight delay in maximum response time during payment spike.
 Notification service (SMS/email) slowed down under heavy load.

Optimization Steps Taken:

Implemented caching for frequently accessed doctor schedules.
 Optimized database queries with indexing.

Increased thread pool size for notification service.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

JMeter load test graph (Response time vs. Users).

Throughput and error rate charts.

(Attach screenshots from your JMeter/Locust reports here.)